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SARDAR PATEL UNIVERSITY

B.C.A. (Sem- 3)

US03SBKA21: Digital Computer Electronics

Date: 14-02-2020

Time: 10.00 am – 12.00 pm

Marks: 35

Q-1 Select the correct option from the following questions. (05)

- 1) An invert gate is also called a _____ gate.
A. NOR B. NOT C. XNOR D. NAND
- 2) The relationship between a function and its binary variables can be represented in _____.
A. Multiplexer B. Decoder C. Encoder D. Truth table
- 3) The NOR gate has two or more input signals. If all inputs are _____, the output is high.
A. low B. high C. both A and B D. none
- 4) Half adder is logic Circuit that adds. _____ Digit at a time.
A. 1 B. 2 C. 3 D. 4
- 5) A register is a group of _____ that work together as a unit.
A. Decoder B. Multiplexer C. Flip-flop D. Gates

Q-2 Do as directed. (ATTEMPT ANY FIVE) (10)

- 1) Describe NAND and OR gate.
- 2) What is quad in k-map?
- 3) Explain De Morgan's Second theorem.
- 4) Write a note on Full Adder.
- 5) Explain sum of product.
- 6) Write a note on Comparator.
- 7) Describe controlled buffer register in brief.
- 8) Explain ring counter in short.

Q-3 A Describe 3*8 line Decoder in detail with truth table and diagram. (05)

B Explain distributive and associative law. (05)

OR**Q-3** A What is k-map? Explain pair and octet with example. (05)

B Explain D flip-flop in detail. (05)

Q-4 A Explain binary adder-subtractor in detail

B Explain 4x1 multiplexer in detail

(05)

(05)

OR

Q-4 A Explain half adder in detail.

B Explain shift left and shift right register.

(05)

(05)

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